

Power BI Basics Assignment

Question 1 : What is Power BI and why is it used in businesses?

Power BI is a business analytics service developed by Microsoft. It is a collection of software services, apps, and connectors that work together to turn unrelated sources of data into coherent, visually immersive, and interactive insights .

Use of Power BI :

Unified Data Visualization: Companies often store data in disparate locations—Excel spreadsheets, cloud services, and on-premises databases like SQL Server. Power BI connects to hundreds of data sources and unifies them into a single "source of truth." This allows businesses to see a holistic view of their operations, from sales figures to inventory levels, in one interactive dashboard.

Real-Time Analytics: In today's fast-paced market, reliance on historical data can be a liability. Power BI supports real-time data streaming, meaning dashboards can update instantaneously as data is generated. This is crucial for businesses that need to monitor manufacturing sensors, social media sentiment, or time-sensitive financial transactions.

Interactive and Actionable Insights: Unlike static reports (like PDFs), Power BI reports are interactive. Users can drill down into specific data points to understand the root cause of a trend. For example, if a sales manager sees a dip in revenue, they can click on that specific month to see which region or product caused the decline. This capability empowers leaders to make evidence-based decisions quickly.

Integration with the Microsoft Ecosystem: For businesses already using Office 365, Power BI is a natural fit. It integrates seamlessly with Excel, SharePoint, and Teams. This integration lowers the learning curve and allows users to embed reports directly into the tools they use daily, ensuring that data insights are part of the regular workflow rather than a separate task.

Cost-Effectiveness and Scalability: Compared to legacy BI tools, Power BI is relatively affordable and scalable. It offers a free desktop version for individual analysis and a scalable monthly subscription model for enterprise sharing, making high-end analytics accessible even to small and medium-sized enterprises.

Question 2 : Name and explain the three main components of Power BI.

Power BI functions through three main components that work in unison to allow users to create, share, and consume business insights. Together, they cover the entire lifecycle of data analysis, from the initial connection to the final presentation on a user's device.

1. Power BI Desktop

This is the **authoring and development tool**. It is a free Windows desktop application where the heavy lifting of data analysis occurs. Analysts use Power BI Desktop to connect to various data sources (like Excel, SQL, or web data), transform and clean that data (using Power Query), and build the data models. Once the data is prepared, this is the environment where users create the actual visual reports—designing charts, graphs, and maps on a canvas. Essentially, if you are building the report, you are using the Desktop.

2. Power BI Service

Often referred to as "Power BI Online," this is the **SaaS (Software as a Service) cloud platform**. Once a report is created in the Desktop application, it is published to the Power BI Service. The Service is primarily used for

collaboration and distribution. Here, users can create dashboards (which aggregate visuals from different reports), share insights with colleagues, and set up scheduled data refreshes so the information remains current. It is the hub where the organization views and interacts with the finished product.

3. Power BI Mobile

This component consists of native mobile applications available for Windows, iOS, and Android devices. Power BI Mobile is the **consumption tool** designed for users on the go. While you cannot create reports here, you can view and interact with the dashboards and reports published to the Power BI Service. It is optimized for touchscreens, allowing executives and field workers to monitor Key Performance Indicators (KPIs) and receive data alerts directly on their phones or tablets, ensuring they have access to critical data anytime, anywhere.

Question 3 : Explain the Power BI workflow

The Power BI workflow is a systematic process of converting raw data into shareable, actionable business insights. It generally follows a linear path from the desktop environment to the cloud, and finally to the end-user's device.

1. Data Preparation and Authoring (Power BI Desktop)

The workflow almost always begins in **Power BI Desktop**. Here, the analyst performs the "ETL" (Extract, Transform, Load) process.

- **Connect:** You start by using "Get Data" to connect to various sources, such as Excel files, SQL databases, or cloud services.
- **Transform:** Using the Power Query Editor, you clean and shape the data—removing errors, renaming columns, and merging tables to make the data usable.
- **Model & Visualize:** Once the data is clean, you create a data model (defining relationships between tables) and build the report by dragging and dropping visual elements like charts, maps, and slicers onto the canvas.

2. Publishing and Collaboration (Power BI Service)

Once the report is complete, the "Publish" button sends it to the **Power BI Service** (cloud). The focus here shifts from creation to distribution.

- **Dashboard Creation:** In the Service, you can pin specific visualizations from your report onto a "Dashboard." A dashboard aggregates the most important metrics from multiple reports into a single-page view.
- **Sharing:** You then share these reports or dashboards with colleagues. This is often done by packaging related content into an "App," which allows for controlled distribution to specific teams or the entire organization.

3. Consumption (Mobile and Web)

Finally, business users consume the insights. They view the reports via a web browser or the **Power BI Mobile** app. They interact with the data—filtering results, drilling down into details, and setting up alerts—to make informed decisions based on the work you started on the Desktop.

Question 4 : List any four data cleaning tasks that can be performed in Power Query.

Here are four common data cleaning tasks we can perform in Power Query to prepare your data for analysis:

1. Removing Rows and Duplicates

Data often comes with unnecessary header rows, footer information, or accidental duplicate entries. Power Query allows you to:

- **Remove Top/Bottom Rows:** Strip out metadata or blank lines that appear before the actual data starts.
- **Remove Duplicates:** we can Scan selected columns to identify and delete identical rows, ensuring your unique counts (like customer IDs) are accurate.

2. Changing Data Types

When data is imported, Power BI might incorrectly identify a date as text or a price as a whole number. We can explicitly define the **Data Type** for each column (e.g., Decimal Number, Date, Text, Boolean). This step is critical because you cannot perform time-intelligence calculations on a date field if Power BI thinks it is just text.

3. Splitting Columns

Sometimes data is grouped too tightly, such as a "Full Name" column containing both first and last names, or an "Address" field containing city and zip code. We can use **Split Column** to separate these values based on a **delimiter** (like a comma or space) or a specific number of characters, allowing you to analyze the components individually.

4. Unpivoting Columns

Data in Excel is often formatted for human reading (wide format), with separate columns for each month (e.g., "Jan Sales," "Feb Sales"). For Power BI analysis, it is better to have a tall, narrow table. **Unpivot** transforms these multiple column headers into a single "Attribute" column (Month) and a "Value" column (Sales Amount), making the data much easier to visualize over time.

Question 5 : Write step by step instructions to load the above dataset

Based on the **given** file , here are the step-by-step instructions to load it into **Power BI Desktop**.

Step 1: Launch Power BI Desktop

We have to Open the Power BI Desktop application on your computer.

Step 2: Get Data

On the **Home** tab of the ribbon at the top of the screen, look for the **Data** section.

1. We have to click on the **Get Data** button.
2. A dropdown menu will appear. Select **CSV**.

Step 3: Locate the File

1. A file explorer window will open. Navigate to the folder where the saved the file is there
2. Select the file and click Open.

Step 4: Preview the Data

Power BI will connect to the file and open a preview window.

1. Verify that the data looks correct. You should see columns like Property_ID, Property_Name, City, Location_Details, Price, and Contact.
2. Check the **Delimiter** dropdown at the top to ensure it is set to **Comma** (Power BI usually detects this automatically).
3. Check that the **Data Type Detection** is set to "Based on the first 200 rows".

Step 5: Load or Transform

At the bottom of the preview window, you will see these buttons.

- **Load:** Click this if the data looks perfect and you want to start building visuals immediately. This will import the data directly into the Report View. (not applicable for his data set)
- **Transform Data:** Click this to open the **Power Query Editor**. *This is recommended for our dataset* because our data has missing values (e.g., in the Price and Area_sqft columns) that you might want to clean up before analyzing.
- After data cleaning we will load the data . Then we

Action: Click **Load** to finish the process. We will see our table names appear in the **Fields** or **Data** pane on the right side of the screen.

Question 6 : Define Data View, Report View, and Model View. Explain the purpose of each view.

Power BI Desktop has three distinct views, accessible from the icon bar on the left side of the application. Each view serves a specific stage in the report development lifecycle.

1. Report View

This is the default view when you open Power BI Desktop. It consists of a blank canvas (our work-in-progress report) surrounded by the **Visualizations** and **Fields** panes.

- **Design & Visualization:** This is where we build the final visual output. We drag and drop fields onto the canvas to create charts, maps, and tables.
- **Layout & Formatting:** We use this view to arrange your visuals, add titles, change colors, and design the aesthetic look of the dashboard that end-users will see.

2. Data View

This view displays your data in a grid format, looking very similar to an Excel spreadsheet or a SQL database table. It shows the raw data rows and columns after they have been loaded into the model.

- **Inspection & Debugging:** It allows you to inspect the actual data underlying your visuals. If a chart looks wrong, you come here to verify if the data rows contain the expected values.
- **Validation:** It is essential for verifying new **Calculated Columns** or **Measures**. We can see exactly how a formula changes the data in the table. We cannot edit data cells here (like in Excel); we can only view them.

3. Model View

This view visualizes our dataset as a schematic diagram (often called an Entity-Relationship or ER diagram). It displays tables as boxes containing their field names, connected by lines representing relationships.

- **Relationship Management:** This is where we define how different tables talk to each other (e.g., linking a "Sales" table to a "Date" table). We configure cardinality (one-to-one, one-to-many) and cross-filter directions here.
- **Schema Organization:** We can create diagrams to group specific tables together, hide technical fields

from the Report View to keep the interface clean, and organize measures into folders.

Question 7 : Discuss the different data sources that Power BI supports.

Power BI supports a vast ecosystem of data sources accessible via the "**Get Data**" feature, allowing users to centralize information from disparate systems. These sources are broadly categorized into:

1. **Files:** The most common sources include Excel workbooks, Text/CSV, XML, JSON, and even PDF files. Power BI can also connect to file folders to combine multiple files automatically.
2. **Databases:** It connects natively to major on-premises and cloud databases like SQL Server, Oracle, IBM Db2, MySQL, PostgreSQL, and SAP HANA.
3. **Online Services (SaaS):** Power BI integrates with popular business applications such as Salesforce, Dynamics 365, Google Analytics, and SharePoint Online Lists.
4. **Azure:** As a Microsoft product, it has deep integration with Azure cloud services, including Azure SQL Database, Synapse Analytics, and Blob Storage.

5. **Web & Other:** Users can scrape data directly from web pages (HTML tables), connect to OData feeds, or use Python and R scripts for custom data extraction.

Question 8 : Split Owner Name to create two new columns as First Name and Last Name

Open Power Query:

- In Power BI Desktop, we have to click the **Home** tab.
- Click on **Transform Data**.

Select the Column:

- Locate and click on the **Owner_Name** column header to select it.

Split the Column:

- Go to the **Home** tab (or **Transform** tab) on the ribbon.
- Click on **Split Column**.
- Select **By Delimiter**.

Configure the Split:

- A dialog box will appear. Ensure the delimiter is set to **Space**.
- **Crucial Step:** Choose **Left-most delimiter**.
 - *Because* If a name is "John Von Neumann", choosing "Left-most" ensures "John" becomes

the First Name and "Von Neumann" becomes the Last Name. If we choose "Each occurrence," you would get three columns.

- Click **OK**.

Rename Columns:

- You will now have two columns (likely named `Owner_Name.1` and `Owner_Name.2`).
- Double-click the header of the first column and rename it to **First Name**.
- Double-click the header of the second column and rename it to **Last Name**.

Close & Apply:

- Go to the **Home** tab and click **Close & Apply** to save changes and return to the report view.

A ^B C First Name	A ^B C Last Name	123
Valid 100% Error 0% Empty 0% 9 distinct, 0 unique	Valid 100% Error 0% Empty 0% 9 distinct, 0 unique	Valid 100% Error 0% Empty 0% 9 distinct, 0 unique
Rahul	Mehta	
Anita	Sharma	
Ramesh	Kumar	
Neha	Verma	
Amit	Joshi	
Suresh	Reddy	
Divya	Iyer	
Manoj	Patil	
Raj	Singh	
Rahul	Mehta	
Anita	Sharma	
Ramesh	Kumar	
Neha	Verma	
Amit	Joshi	
Suresh	Reddy	
Divya	Iyer	
Manoj	Patil	
Raj	Singh	
Rahul	Mehta	
Anita	Sharma	
Ramesh	Kumar	

